EXNO: 4 -Strategize impedance matching with characteristic impedance and load impedance through the measurement of

a) Location of the stub

b) Length of the stub

Test Case 1:

Optimize impedance matching using single stub matching by finding the location and length of the stub for the values of load impedance of 100 30o to 350 30o keeping the characteristic impedance Zo= 60 Ω constant .

AIM: To optimize impedance matching using single stub matching by finding the location and length of the stub for the values of load impedance of 100 30o to 350 30o keeping the characteristic impedance Zo= 60 Ω constant .

SOFTWAREREQUIRED:

SCILAB

PROCEDURE:

● Open SCILAB software

● Open file-new

● Put the input values

● Type the program

● Execute the program

● End the program

PROGRAM:

```
clear; clear; clc;

ZR=100;Zo=600;f=100*(10^6);

lo=300/(f*(10^-6)); //lo=wavelength Ls=(lo/(2*%pi))*(atan(sqrt(ZR/Zo)));

printf("-Point of attachment = %f cms\n",round(Ls*(10^2)*10)/10)

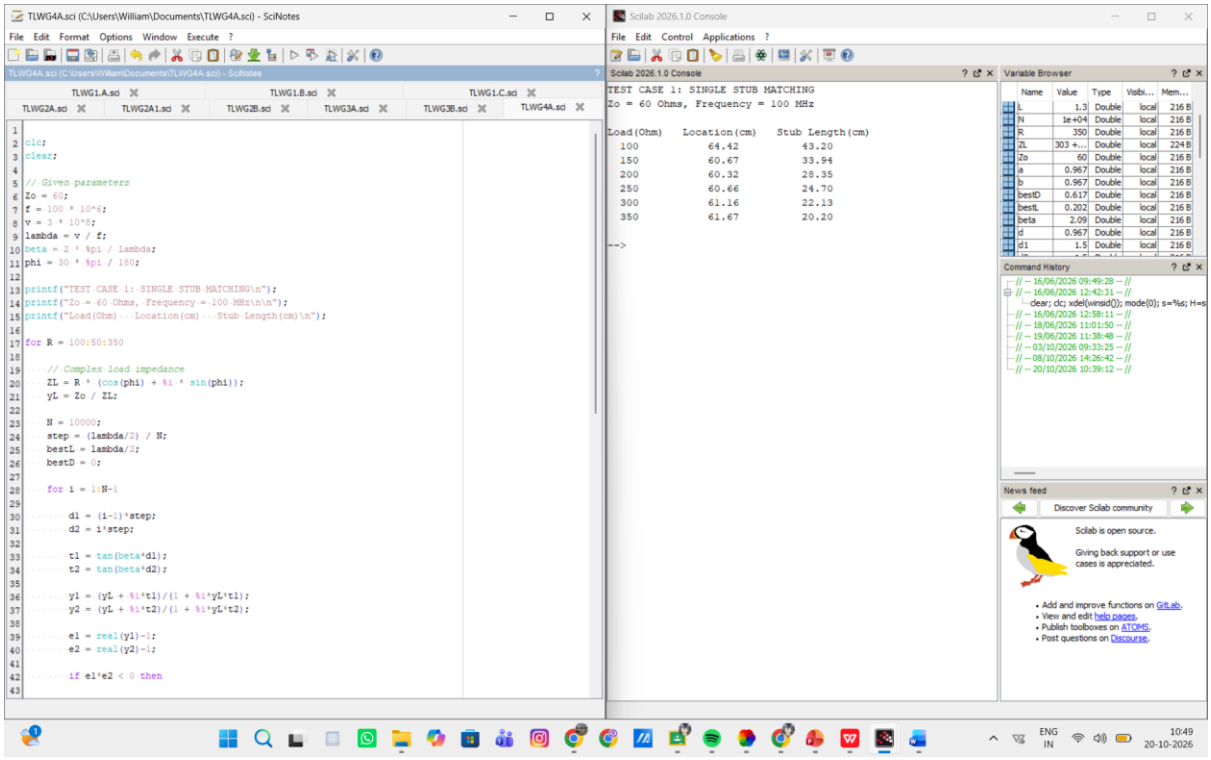
Lt=(lo/(2*%pi))*(%pi+(atan((sqrt(ZR*Zo))/(ZR-Zo))));

printf("-Length of the short circuited stub = %f cms",round(Lt*(10^2)));
```

OUTPUT:

Variables ZR;

Constants Zo = 60Ω; f=100*(10^6);



S. No.	Load impedance \(Z_L\)	Stub location (cm)	Stub length (cm)
1	\(100\angle30^\circ\,\Omega\)	64.42	43.20
2	\(150\angle30^\circ\,\Omega\)	60.67	33.94
3	\(200\angle30^\circ\,\Omega\)	60.32	28.35
4	\(250\angle30^\circ\,\Omega\)	60.66	24.70
5	\(300\angle30^\circ\,\Omega\)	61.16	22.13
6	\(350\angle30^\circ\,\Omega\)	61.67	20.20

RESULT:

Impedance matching using single stub matching is optimized by calculating the location and length of the stub f or varying values of load impedance keeping the characteristic impedance Zo= 60 Ω constant

Test Case 2:

Optimize impedance matching using single stub matching by finding the location and length of the stub for the values of characteristic impedance ranging from 600Ω to 1800 Ω. Keeping the load impedance ZR= 100 30o constant .

AIM: To optimize impedance matching using single stub matching by finding the location and length of the stub for the values of characteristic impedance ranging from 600Ω to 1800 Ω. Keeping the load impedance ZR= 100 30o constant .

SOFTWAREREQUIRED:

SCILAB

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clear; clear; clc;
ZR=100;Zo=600;f=100*(10^6);
lo=300/(f*(10^-6)); //lo=wavelength Ls=(lo/(2*%pi))*(atan(sqrt(ZR/Zo)));
printf("-Point of attachment = %f cms\n",round(Ls*(10^2)*10)/10)
Lt=(lo/(2*%pi))*(%pi+(atan((sqrt(ZR*Zo))/(ZR-Zo))));
printf("-Length of the short circuited stub = %f cms",round(Lt*(10^2)));
```

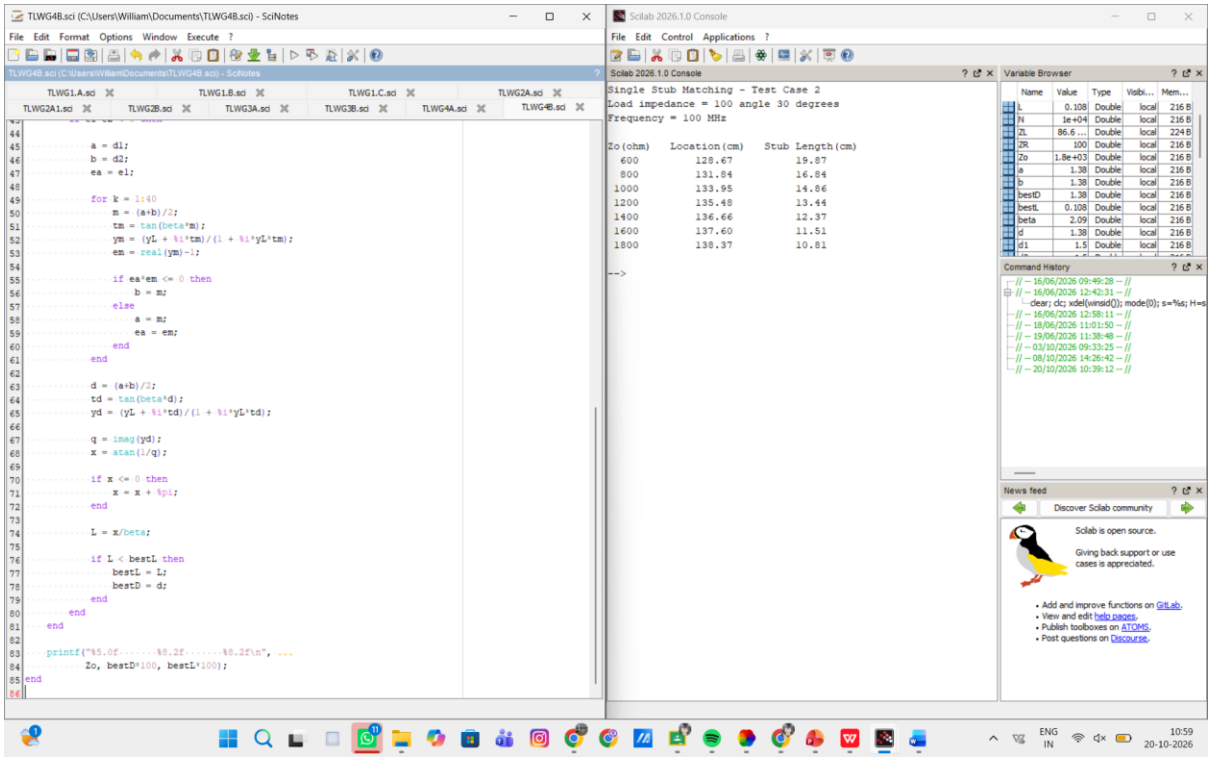
OUTPUT:

Variables Z0;

Constants ZR= 100 30oΩ;

f=100*(10^6);

S. No.	\(Z_0\) (Ω)	Stub location (cm)	Stub length (cm)
1	600	128.67	19.87
2	800	131.84	16.84
3	1000	133.95	14.86
4	1200	135.48	13.44
5	1400	136.66	12.37
6	1600	137.60	11.51
7	1800	138.37	10.81



RESULT:

Impedance matching using single stub matching is optimized by calculating the location and length of the stub for varying values of characteristic impedance keeping the load impedance constant .